UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION - SEMESTER I (2021/2022)
CSC1013 – INTRODUCTION TO COMPUTER SCIENCE AND PROGRAMMING (3 Credits)

Time Allowed: **TWO (02)** Hours　　　　Answer **ALL** Questions

Instructions

- There are **Four (04)** questions **on Three (03)** pages.
- All questions carry equal marks.
- Clearly indicate the Question number and subsection number in each answer. E.g.: 2 (b)
- Make your answers concise and succinct.

PART A

1.

a. Briefly explain how magnetic storage is used to store binary data. **[02 marks]**

b. Using a diagram briefly explain how raw image data is stored in a computer. **[03 marks]**
Explain the concepts of lossy and lossless image compression. **[02 marks]**

c. Briefly explain the technical reason why Unicode was developed to replace ASCII as the mechanism for storing character data in a computer system. **[02 marks]**

d. Select the most appropriate answer. (Write the question number and letter of the corresponding answer.) **[16 marks]**

i. A byte is a signed integer data type in Java. What is the range of values stored in a byte?
(a) 0 to 256　(b) -128 to +128　(c) -127 to 127　(d) -128 to 127

ii. What is the binary representation of the decimal value 155?
(a) 10110101　(b) 10011011　(c) 0110100　(d) 11011001

iii. What is the decimal representation of the binary value 01100101?
(a) 101　(b) 100　(c) 166　(d) 165

iv. What is the octal representation of the hexadecimal value A023?
(a) 40445　(b) 100023　(c) 120043　(d) 40443

v. Which of the following is the equivalent binary value for decimal value 0.09375?
(a) 0.0001　(b) 0.00011　(c) 0.00110　(d) 0.0001101

vi. Which of the following logic gates provides an output of 1 **when and only when** both inputs are 0?
(a) XOR　(b) XNOR　(c) NOR　(d) NAND

vii. Which of the following is the correct simplification for $A + (A.B)$ using the absorption law?
(a) A　(b) A + B　(c) B　(d) $A.B$

viii. Which of the following is the largest decimal number that can be represented in Signed Fixed-point <8,2> number representation?

(a) +63.75	(b) +64.25	(c) +32.75	(d) +31.75

2.

a. Represent 127.0625_{10} using IEEE 754 32-bit representation. **[04 marks]**

b. Perform the following binary arithmetic operations. Clearly indicate the steps.

i. 00011010 – 00101100 Using 2's complement **[01 mark]**

ii. 00101100 / 00001000 **[01 mark]**

c. Perform the arithmetic operation $44_{10} - 26_{10}$ using BCD. Use 9's complement. **[03 mark]**

d. Simplify the product of sums expression $F(ABC) = \prod(M_1, M_4, M_5, M_6, M_7)$ using Boolean algebraic laws. **[06 marks]**

e. You are required to design a logic circuit that turns on and off an automatic sprinkler system depending on the moisture level in the soil. The moisture sensor has two binary outputs. It has been programmed to provide a binary 1 when the moisture level reaches the required level, else the output is binary 0 on the 1st output. The moisture sensor provides a binary 1 when the moisture level reaches the minimum level, else the output is binary 0 on the 2nd output. The sprinkler (a water spray) needs to turn on when the moisture level reaches the minimum level and turn off when the moisture level reaches the required level. The sprinkler is turned on by an electronically controlled water tap. The tap provides an output of binary 1 if the tap is in the open state and binary 0 if it is in the closed state. The tap opens on binary 1 and closes on binary 0. Design a minimalistic logic circuit (least number of logic gates) which takes two inputs from the moisture sensor and one input from the tap state and provides an output to turn the tap on and off as required. Draw the logic circuit. *Hint: Use a Truth Table and a Karnaugh Map* **[08 marks]**

f. You have been provided with a 1-to-8 demultiplexer. How many control signals are needed for this demultiplexer and clearly explain its functionality. **[02 marks]**

PART B

3.

a. Explain three significant reasons to have multiple data types in programming languages. **[06 marks]**

b. Describe the difference between parameters and arguments with a suitable C or Python language example. **[04 marks]**

c. Develop a code snippet to demonstrate 'Initialization of variables simplifies the flow control of programmes'. (You can use C or Python as the programming) **[03 marks]**

d. Develop a flowchart (with standard symbols) to display all the numbers having the digit 9 for given boundaries (lower and upper limits, both inclusive). **[07 marks]**

Example:
Input: lower = 1, upper = 100
Output: 9, 19, 29, 39, 49, 59, 69, 79, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99

e. List three significant reasons to use functions in programming. **[03 marks]**

f. Describe two main differences between Structured programming and Procedural programming **[02 marks]**

4.

a. Write the outputs of the following Python snippets. **[10 marks]**

i)

```
def Display(str):
        m=""
        for i in range(0,len(str)):
                if(str[i].isupper()):
                        m=m+str[i].lower()
                elif str[i].islower():
                        m=m+str[i].upper()
                else:
                        if i%2==0:
                                m=m+str[i-1]
                        else:
                                m=m+"#"
        print(m)
Display('Fun@Python3.0')
```

ii)

```
def display(x=2, y=3):
        x = x+y
        y  += 2
        print(x,y)
display()
display(5, 1)
display(9)
```

2+3

x=5
y=5

iii)

```
def increment(n):
        n.append([4])
        return n
        L=[1,2,3]
        M=increment(L)
        print(L, M)
```

[1,2,3,[4]

b. Describe two differences between Python lists and tuples. **[03 marks]**

c. Online banking systems need an account name verification mechanism. However, personal name identification is not straightforward as names can be written in many forms. **[12 marks]**

E.g. Account name in the bank database: John Eric Wilson
Alternative name patterns could be filled in web forms: JE Wilson, J Erik Wilson, John E. Wilson, Wilson JE, Eric Wilson … Also, there can be typos while entering the name.

Develop a complete Python programme to check similarities of names in the bank database (John Eric Wilson), and web form filled names (e.g. JE Wilson).